P2P Architectural diagram

Current Risk Summary report

Fri Mar 13 2026 06:48:43 GMT+0000 (Coordinated Universal Time)

Project description:
Illustrating the peer-to-peer chat system with a focus on trust boundaries and data flow

Filtered by: *No filters*

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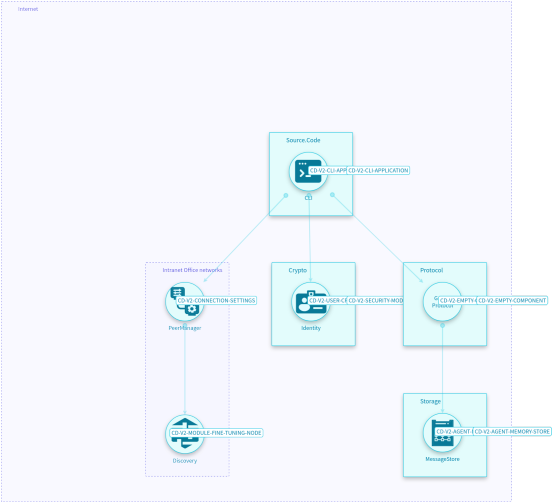
Owner: mister man, mister man, Muhammad Zaeem Nawaz

Workflow state: Draft

Tags: *No tags*



Content menu

Current risk summary

Components

Accepted Risks

Current Risks

- CLI
- Crypto
- Discovery
- Identity
- MessageStore
- PeerManager
- Source.Code
- Storage

Current Risk summary

Inherent risk description: The Inherent Risk before countermeasures were applied.
- **Risk Rating:** 63.80303192138672% High

The Current Risk description (the risk we are at now): The Current Risk is based on the current implementation status of the countermeasures and test results.
- **Risk Rating:** 63.80303192138672% High

Projected Risk description: The Projected Risk is the level of risk that would be reached should the required countermeasures be implemented.
- **Risk Rating:** 63.80303192138672% High

Components

- CLI

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Sent from component
  - **Customer Data: How is it handled by this component?** Stored
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component handle personally identifiable information from citizens of the European Union?** Yes
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Stored
  - **Protected Health Information: How is it handled by this component?** Processed

- Crypto

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Stored
  - **Customer Data: How is it handled by this component?** Stored
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Processed
  - **Protected Health Information: How is it handled by this component?** Processed

- Discovery

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Sent from component
  - **Customer Data: How is it handled by this component?** Processed
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Processed
  - **Protected Health Information: How is it handled by this component?** Processed
  - **Which MAESTRO layers best describe the role and interactions of this component?** MAESTRO Cross-Layer
  - **Which MAESTRO layers best describe the role and interactions of this component?** MAESTRO Layer 5 - Evaluation and observability

- Gossip Protocol

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Stored
  - **Customer Data: How is it handled by this component?** Sent from component
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Received by component
  - **Protected Health Information: How is it handled by this component?** Stored

- Identity

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Stored
  - **Customer Data: How is it handled by this component?** Stored
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Stored
  - **Protected Health Information: How is it handled by this component?** Received by component

- MessageStore

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Stored
  - **Customer Data: How is it handled by this component?** Stored
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component handle personally identifiable information from citizens of the European Union?** Yes
  - **Does this component have to be CCPA-compliant?** Yes
  - **Personally Identifiable Information: How is it handled by this component?** Stored
  - **Protected Health Information: How is it handled by this component?** Processed
  - **Which MAESTRO layers best describe the role and interactions of this component?** MAESTRO Layer 2 - Data Operations
  - **Which of the following CCPA rights do you want this software component to be compliant with?** Right to Know
  - **Which of the following CCPA rights do you want this software component to be compliant with?** Right to Correct
  - **Which of the following CCPA rights do you want this software component to be compliant with?** Right to Delete
  - **Which of the following CCPA rights do you want this software component to be compliant with?** Right to Non-Discrimination

- **Which of the following CCPA rights do you want this software component to be compliant with?** Right to Limit

- PeerManager

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Stored
  - **Customer Data: How is it handled by this component?** Processed
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Processed
  - **Protected Health Information: How is it handled by this component?** Processed

- Protocol

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Stored
  - **Customer Data: How is it handled by this component?** Stored
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Stored
  - **Protected Health Information: How is it handled by this component?** Processed

- Source.Code

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Sent from component
  - **Customer Data: How is it handled by this component?** Stored
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Processed
  - **Protected Health Information: How is it handled by this component?** Processed

- Storage

  Model questionnaire information:
  - **Credit Card Data: How is it handled by this component?** Processed
  - **Customer Data: How is it handled by this component?** Processed
  - **Does this component handle personally identifiable information from citizens of the European Union?** No
  - **Does this component have to be CCPA-compliant?** No
  - **Personally Identifiable Information: How is it handled by this component?** Stored
  - **Protected Health Information: How is it handled by this component?** Stored
  - **Which MAESTRO layers best describe the role and interactions of this component?** MAESTRO Layer 2 - Data Operations

Accepted Risks

No data

Current Risks

Component: CLI

Use case: Information Disclosure

CRT1. Threat name: Compromised system due to misconfiguration or vulnerable credential management
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR1. Countermeasure name:** Comprehensive security configuration and credential management
  - **Status:** RECOMMENDED

Use case: Elevation of Privilege

CRT2. Threat name: Data security flaws for insecure deserialization
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR2. Countermeasure name:** Implement secure deserialization practices
  - **Status:** RECOMMENDED

CRT3. Threat name: Unprivileged access via vulnerable dependencies
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR3. Countermeasure name:** Establish regular dependency scanning processes
  - **Status:** RECOMMENDED

Use case: Tampering

CRT4. Threat name: Input manipulation and injections attacks
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR4. Countermeasure name:** Implement comprehensive input validation and sanitization
  - **Status:** RECOMMENDED

CRT5. Threat name: Tampering of URLs into unvalidated redirects and forwards
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR5. Countermeasure name:** Implement URL validation and sanitization
  - **Status:** RECOMMENDED

Component: Crypto

Use case: Tampering

CRT6. Threat name: Attacker exploit misconfiguration and vulnerable third-party
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR6. Countermeasure name:** Comprehensive configuration hardening and dependency auditing
  - **Status:** RECOMMENDED

CRT7. Threat name: Attackers execute injection attacks
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR7. Countermeasure name:** Use parameterized queries and input validation
  - **Status:** RECOMMENDED

Use case: Spoofing

CRT8. Threat name: Attackers gain elevated privileges or unauthorized access
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR8. Countermeasure name:** Implement multi-layered security for authentication and access control
  - **Status:** RECOMMENDED

Use case: Information Disclosure

CRT9. Threat name: Attackers gather useful information from inadequate error handling
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR9. Countermeasure name:** Implement generic error messages and proper exception handling
  - **Status:** RECOMMENDED

CRT10. Threat name: Attackers take advantage of insecure communication channels and unprotected sessions
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR10. Countermeasure name:** Implement comprehensive secure communication and session management protocols
  - **Status:** RECOMMENDED

CRT11. Threat name: Attackers take advantage of weak encryption of sensitive data
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR11. Countermeasure name:** Implement and maintain advanced encryption standards with effective key management
  - **Status:** RECOMMENDED

Use case: Denial of Service

CRT12. Threat name: Attackers perform a Denial of Service (DoS)
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR12. Countermeasure name:** Implement rate limiting and resource allocation
  - **Status:** RECOMMENDED

Use case: Repudiation

CRT13. Threat name: Lack of evidences of misuse due to insufficient auditing and logging and poor log protection
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR13. Countermeasure name:** Implement comprehensive logging and monitoring with log integrity measures
  - **Status:** RECOMMENDED

Component: Discovery

Use case: MAESTRO Layer 5 - Evaluation and observability

CRT14. Threat name: Compromised observability tools
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR14. Countermeasure name:** Secure monitoring tools with code reviews and integrity checks
  - **Status:** RECOMMENDED

CRT15. Threat name: Data leakage through observability
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR15. Countermeasure name:** Apply strict access control and data minimization in logs and dashboards

- **Status:** RECOMMENDED

CRT16. Threat name: Denial of service on evaluation infrastructure
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR16. Countermeasure name:** Deploy redundancy, rate limiting and DoS protection
  - **Status:** RECOMMENDED

CRT17. Threat name: Evasion of detection
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR17. Countermeasure name:** Enhance detection with anomaly monitoring and adversarial testing
  - **Status:** RECOMMENDED

CRT18. Threat name: Manipulation of evaluation metrics
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR18. Countermeasure name:** Use robust benchmarks and validate test datasets
  - **Status:** RECOMMENDED

CRT19. Threat name: Poisoning observability data
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR19. Countermeasure name:** Validate and cross-check observability data sources for integrity
  - **Status:** RECOMMENDED

Use case: MAESTRO Cross-Layer

CRT20. Threat name: Data leakage
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR20. Countermeasure name:** Apply strict data governance and isolation
  - **Status:** RECOMMENDED

CRT21. Threat name: Goal misalignment cascades
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR21. Countermeasure name:** Implement multi-layer validation and oversight
  - **Status:** RECOMMENDED

CRT22. Threat name: Lateral movement
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR22. Countermeasure name:** Apply network segmentation and zero-trust controls
  - **Status:** RECOMMENDED

CRT23. Threat name: Privilege escalation
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR23. Countermeasure name:** Enforce least privilege and strong identity controls
  - **Status:** RECOMMENDED

CRT24. Threat name: Supply chain attacks
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR24. Countermeasure name:** Enforce end-to-end supply chain security
  - **Status:** RECOMMENDED

Use case: Tampering

CRT25. Threat name: Introduce malicious or untrusted data into fine-tuning node
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR25. Countermeasure name:** Verify and constrain fine-tuning datasets
  - **Status:** RECOMMENDED

CRT26. Threat name: Tamper with fine-tuning artifacts and outputs
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR26. Countermeasure name:** Protect artifacts and limit exposed internals
  - **Status:** RECOMMENDED

Use case: Information Disclosure

CRT27. Threat name: Leak or extract sensitive data from fine-tuning node
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR27. Countermeasure name:** Monitor and restrict model outputs
  - **Status:** RECOMMENDED
- **CR28. Countermeasure name:** Apply privacy-preserving fine-tuning and pipeline hardening
  - **Status:** RECOMMENDED

Use case: Elevation of Privilege

CRT28. Threat name: Weaken safety alignment of tuned node
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR29. Countermeasure name:** Test and reinforce alignment after fine-tuning
  - **Status:** RECOMMENDED

Component: Identity

Use case: Spoofing

CRT29. Threat name: Attackers compromise user accounts by authentication attacks
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR30. Countermeasure name:** Implement secure password hashing and multi-factor authentication
  - **Status:** RECOMMENDED
- **CR31. Countermeasure name:** Implement account lockout policy and CAPTCHA
  - **Status:** RECOMMENDED

Use case: Information Disclosure

CRT30. Threat name: Attackers exploit misconfiguration and supply chain vulnerabilities
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR32. Countermeasure name:** Harden system configurations and conduct third-party security assessments
  - **Status:** RECOMMENDED

CRT31. Threat name: Attackers gather useful information from inadequate error handling
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR33. Countermeasure name:** Implement secure error handling
  - **Status:** RECOMMENDED

CRT32. Threat name: Attackers take advantage of insufficient sata protection and encryption for user credentials

- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR34. Countermeasure name:** Implement strong encryption for data at rest
  - **Status:** RECOMMENDED
- **CR35. Countermeasure name:** Apply data masking and tokenization
  - **Status:** RECOMMENDED
- **CR36. Countermeasure name:** Secure data transmission
  - **Status:** RECOMMENDED

Use case: Elevation of Privilege

CRT33. Threat name: Attackers gain elevated privileges through credential misuse
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR37. Countermeasure name:** Enforce principle of least privilege
  - **Status:** RECOMMENDED

Use case: Denial of Service

CRT34. Threat name: Attackers perform Denial of Service through authentication systems
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR38. Countermeasure name:** Implement rate limiting
  - **Status:** RECOMMENDED

Use case: Tampering

CRT35. Threat name: Attackers perform injection attacks and unathorized data modification
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR39. Countermeasure name:** Use prepared statements with parameterized queries
  - **Status:** RECOMMENDED
- **CR40. Countermeasure name:** Implement input validation
  - **Status:** RECOMMENDED

Use case: Repudiation

CRT36. Threat name: Repudiation of authentication actions
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR41. Countermeasure name:** Implement comprehensive logging
  - **Status:** RECOMMENDED

Component: MessageStore

Use case: MAESTRO Layer 2 - Data Operations

CRT37. Threat name: Compromised RAG pipelines
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR42. Countermeasure name:** Enforce input sanitization and process isolation (sandboxing)
  - **Status:** RECOMMENDED

CRT38. Threat name: Data exfiltration
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR43. Countermeasure name:** Apply comprehensive encryption and least privilege access control

- **Status:** RECOMMENDED

CRT39. Threat name: Data poisoning
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR44. Countermeasure name:** Implement input validation and anomaly detection
  - **Status:** RECOMMENDED

CRT40. Threat name: Data tampering
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR45. Countermeasure name:** Use cryptographic integrity verification
  - **Status:** RECOMMENDED

CRT41. Threat name: Denial of service on data infrastructure
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR46. Countermeasure name:** Implement a redundant architecture and DDoS mitigation
  - **Status:** RECOMMENDED

Use case: Tampering

CRT42. Threat name: Compromise memory through abnormal growth or access patterns
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR47. Countermeasure name:** Monitor and audit memory operations
  - **Status:** RECOMMENDED

CRT43. Threat name: Exploit uncontrolled or adversarial memory inputs
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR48. Countermeasure name:** Filter and control external inputs
  - **Status:** RECOMMENDED
- **CR49. Countermeasure name:** Define trust boundaries and permissions
  - **Status:** RECOMMENDED

CRT44. Threat name: Poison or manipulate stored memory
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR50. Countermeasure name:** Apply provenance and source attribution
  - **Status:** RECOMMENDED
- **CR51. Countermeasure name:** Validate and verify memory integrity
  - **Status:** RECOMMENDED
- **CR52. Countermeasure name:** Restrict and monitor external memory writes
  - **Status:** RECOMMENDED

Use case: Information Disclosure

CRT45. Threat name: Leak or misuse memory across sessions and users
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR53. Countermeasure name:** Configure retention and user oversight
  - **Status:** RECOMMENDED
- **CR54. Countermeasure name:** Enforce isolation and access control
  - **Status:** RECOMMENDED

Component: PeerManager

Use case: Information Disclosure

CRT46. Threat name: Attackers take advantage of improperly secured system configurations and the handling of input data
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR55. Countermeasure name:** Implement comprehensive security configuration and validation framework
  - **Status:** RECOMMENDED

CRT47. Threat name: Inadequate error handling exposing sensitive information
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR56. Countermeasure name:** Implement secure error handling practices
  - **Status:** RECOMMENDED

CRT48. Threat name: Securing communication channels to prevent unauthorized access and data exposure
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR57. Countermeasure name:** Comprehensive encryption and secure configuration
  - **Status:** RECOMMENDED

CRT49. Threat name: Side-channel attacks
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR58. Countermeasure name:** Implement cryptographic best practices
  - **Status:** RECOMMENDED

Use case: Denial of Service

CRT50. Threat name: Denial of Service and resource exhaustion
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR59. Countermeasure name:** Implement comprehensive DoS protection strategies
  - **Status:** RECOMMENDED

Use case: Elevation of Privilege

CRT51. Threat name: Elevation of privilege through connection misconfiguration
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR60. Countermeasure name:** Enforce least privilege and segregation of duties
  - **Status:** RECOMMENDED

Use case: Repudiation

CRT52. Threat name: Repudiation of actions over connections
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR61. Countermeasure name:** Secure logging and monitoring
  - **Status:** RECOMMENDED

Component: Source.Code

Use case: Information Disclosure

CRT53. Threat name: Compromised system due to misconfiguration or vulnerable credential management
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR62. Countermeasure name:** Comprehensive security configuration and credential management

- **Status:** RECOMMENDED

Use case: Elevation of Privilege

CRT54. Threat name: Data security flaws for insecure deserialization
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR63. Countermeasure name:** Implement secure deserialization practices
  - **Status:** RECOMMENDED

CRT55. Threat name: Unprivileged access via vulnerable dependencies
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR64. Countermeasure name:** Establish regular dependency scanning processes
  - **Status:** RECOMMENDED

Use case: Tampering

CRT56. Threat name: Input manipulation and injections attacks
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR65. Countermeasure name:** Implement comprehensive input validation and sanitization
  - **Status:** RECOMMENDED

CRT57. Threat name: Tampering of URLs into unvalidated redirects and forwards
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR66. Countermeasure name:** Implement URL validation and sanitization
  - **Status:** RECOMMENDED

Component: Storage

Use case: MAESTRO Layer 2 - Data Operations

CRT58. Threat name: Compromised RAG pipelines
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR67. Countermeasure name:** Enforce input sanitization and process isolation (sandboxing)
  - **Status:** RECOMMENDED

CRT59. Threat name: Data exfiltration
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR68. Countermeasure name:** Apply comprehensive encryption and least privilege access control
  - **Status:** RECOMMENDED

CRT60. Threat name: Data poisoning
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR69. Countermeasure name:** Implement input validation and anomaly detection
  - **Status:** RECOMMENDED

CRT61. Threat name: Data tampering
- **Inherent risk:** Medium
- **Current risk:** Medium
- **Projected risk:** Medium
- **State:** Exposed
- **CR70. Countermeasure name:** Use cryptographic integrity verification
  - **Status:** RECOMMENDED

CRT62. Threat name: Denial of service on data infrastructure
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR71. Countermeasure name:** Implement a redundant architecture and DDoS mitigation
  - **Status:** RECOMMENDED

Use case: Tampering

CRT63. Threat name: Compromise memory through abnormal growth or access patterns
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR72. Countermeasure name:** Monitor and audit memory operations
  - **Status:** RECOMMENDED

CRT64. Threat name: Exploit uncontrolled or adversarial memory inputs
- **Inherent risk:** High
- **Current risk:** High
- **Projected risk:** High
- **State:** Exposed
- **CR73. Countermeasure name:** Filter and control external inputs
  - **Status:** RECOMMENDED
- **CR74. Countermeasure name:** Define trust boundaries and permissions
  - **Status:** RECOMMENDED

CRT65. Threat name: Poison or manipulate stored memory
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR75. Countermeasure name:** Apply provenance and source attribution
  - **Status:** RECOMMENDED
- **CR76. Countermeasure name:** Validate and verify memory integrity
  - **Status:** RECOMMENDED
- **CR77. Countermeasure name:** Restrict and monitor external memory writes
  - **Status:** RECOMMENDED

Use case: Information Disclosure

CRT66. Threat name: Leak or misuse memory across sessions and users
- **Inherent risk:** Critical
- **Current risk:** Critical
- **Projected risk:** Critical
- **State:** Exposed
- **CR78. Countermeasure name:** Configure retention and user oversight
  - **Status:** RECOMMENDED
- **CR79. Countermeasure name:** Enforce isolation and access control
  - **Status:** RECOMMENDED

End of Current Risk Report